1. Score equation and estimator

Your approximate score is

$$\frac{\partial \ell(\sigma^2)}{\partial \sigma^2} \approx -\frac{N_T - 1}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{n=1}^{N_T-1} \frac{(x_{l_{n+1}} - x_{l_n})^2}{l_{n+1} - l_n}.$$

Setting this equal to zero and multiplying by $2\sigma^4$,

$$-(N_T - 1)\sigma^2 + \sum_{n=1}^{N_T-1} \frac{(x_{l_{n+1}} - x_{l_n})^2}{l_{n+1} - l_n} = 0.$$

Hence the (approximate) MLE is

$$\boxed{\widehat{\sigma}^2 = \frac{1}{N_T - 1} \sum_{n=1}^{N_T-1} \frac{(x_{l_{n+1}} - x_{l_n})^2}{l_{n+1} - l_n}}$$

This is a **mean of squared record increments normalized by their waiting times**, i.e. a natural estimator of the variance rate.

2. Second-order condition

Your second derivative is

$$\frac{\partial^2 \ell(\sigma^2)}{\partial(\sigma^2)^2} \approx \frac{N_T - 1}{2\sigma^4} - \sum_{n=1}^{N_T-1} \frac{(x_{l_{n+1}} - x_{l_n})^2}{(l_{n+1} - l_n)\sigma^6}.$$

Evaluating it at $\widehat{\sigma}^2$ and using the score equation,

$$\left.\frac{\partial^2 \ell(\sigma^2)}{\partial(\sigma^2)^2}\right|_{\widehat{\sigma}^2} = -\frac{N_T - 1}{2(\widehat{\sigma}^2)^2} < 0,$$

confirming that $\widehat{\sigma}^2$ is indeed a (local) maximum.

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3. Fisher information and variance

Under the model,

$$\mathbb{E}\left[\frac{(x_{l_{n+1}} - x_{l_n})^2}{l_{n+1} - l_n}\right] = \sigma^2,$$

so

$$\mathbb{E}\left[\frac{\partial^2 \ell(\sigma^2)}{\partial(\sigma^2)^2}\right] = -\frac{N_T - 1}{2\sigma^4}.$$

Thus the Fisher information is

$$\mathcal{I}(\sigma^2) = \frac{N_T - 1}{2\sigma^4},$$

and the asymptotic variance of the estimator is

$$\boxed{\mathrm{Var}(\widehat{\sigma}^2) = \frac{2\sigma^4}{N_T - 1}}.$$

In empirical applications,

$$\mathrm{Var}(\widehat{\sigma}^2) \approx \frac{2(\widehat{\sigma}^2)^2}{N_T - 1}.$$

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4. Asymptotic distribution

Provided $N_T \to \infty$,

$$\sqrt{N_T - 1}\,(\widehat{\sigma}^2 - \sigma^2) \xrightarrow{d} \mathcal{N}(0,\, 2\sigma^4),$$

or equivalently,

or equivalently,

$$\widehat{\sigma}^2 \overset{\cdot}{\sim} \mathcal{N}\left(\sigma^2,\, \frac{2\sigma^4}{N_T - 1}\right).$$

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5. Interpretation (important for reviewers)

- Only **record increments** are used, so the effective sample size is $N_T - 1$, not T.
- The estimator is **consistent but less efficient** than the full-data estimator.
- Random record spacings enter **only through normalization**, and their randomness does *not* alter the variance rate.

Final takeaway

$$\widehat{\sigma}^2 = \frac{1}{N_T - 1} \sum_{n=1}^{N_T - 1} \frac{(x_{l_{n+1}} - x_{l_n})^2}{l_{n+1} - l_n}, \qquad \mathrm{Var}(\widehat{\sigma}^2) = \frac{2\sigma^4}{N_T - 1}.$$